

A Proposal for a Master's Thesis

To be Entitled

Surveillance of HIV Pre-Exposure Prophylaxis Provision within the SHI System in Germany
(PrEP-Surv)

by

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1 Title

Surveillance of HIV Pre-Exposure Prophylaxis Provision within the SHI System in Germany (PrEP-Surv)

2 Abstract

This project aims to enhance HIV prevention efforts in Germany by leveraging statutory health insurance (SHI) to monitor and improve the provision of HIV pre-exposure prophylaxis (PrEP). Through advanced data handling methods, integration of Fast Healthcare Interoperability Resources (FHIR), and Medical Information Objects (MIO), this study will address critical gaps in HIV monitoring and PrEP service delivery. The project emphasizes developing a user-friendly, interoperable digital platform for healthcare providers and patients, ensuring efficient data management and accessibility.

3 Introduction

The prep Surv is an initiative organized by the German Federal Ministry of Health (BMG) and associated with Statutory health insurance (STI) for the surveillance and coverage of HIV pre-exposure among the HIV-negative population. The coverage of this initiative includes, administration of the drug [tenofovir disoproxil + emtricitabine, TDF/FTC], counseling, the supply of medication and the necessary precautions including further examination. This project constitutes the association of the previous project 'Evaluation of the introduction of HIV pre-exposure prophylaxis as a benefit of statutory health insurance'. The start of Prev-Surv is the projection of the results of the prEP Evaluation. The preliminary focus of this project is to monitor (according to the guidelines of EU-prEP monitoring) the development of STI and HIV among the PreP users in Germany. The process needs the requirement for huge data influx from various health entities and communal bodies. The data retrieval and provisional service is done in terms of three modules (1) Secondary data from pharmacies, health insurance companies and from the Health data lab (FDZ) (2) Strengthening the provisional service of prEP surv at German HIV specialty care centers(in the dāgna network) by collecting the prEP service data (3) Involvement of communal bodies to address the need for prEP service as well to understand barriers for future penetration in the space of HIV-monitoring.

The digitalization of the objective service is to create a medical information object (MIO) for the prEP users. The Medical Information Object (MIO) is a standardized data structure in the electronic patient record (ePA) in Germany. In the context of the legal mandate defined by the KBV, the term MIO is used for all the digital information modules that are used to document medical data in a defined data format. MOI is a digital module which works with data and is able to take data structured in particular formats for further analysis. In short, Medical Information Objects are packages of data in an ePA. There can be multiple MIOs for a single patient, ECG MIO, Blood MIO, Medication MIO etc. In short, an MOI is a readable and editable digital information module which works with the property of interoperability in every semantic and syntactic way.

4 Objectives

Traditional bioinformatics and medical informatics applications lack the GUI (Graphical User Interface) versatility since the architecture is single-tranche based. The primary mode of kernel manipulation is via terminal --Command Line Interfacing (CLI). The element of GUI is necessary to extract, manipulate, derive

information in a structured and efficient way from the electronic health records. The Medical Information Object (MOI) is one such way to have a standardized template which satisfies the property of multi-tranche-based GUI. The scalability of bio/medical informatics to reach potential end users to solve large-scale health problems depends on the ability to have such standardized templates of GUI frameworks. The productive computer application is composed of three-tranche GUI architecture namely (1) Structured Elements (2) Interactive Elements (3) Graphical control Elements. Bioinformatics applications are based solely on the Graphical control Elements which hold data-input-output functions focusing on Command Line Interfacing (CLI) neglecting the other two elements. Even though the preliminary objective of bio/medical informatics tools is to run the backend application in a fast and cache-friendly manner to save the access-memory, the importance of replacing esoteric data interfaces in the terminal to make the end-user easy to use should be considered. Medical informatics objects (MOI) are bootstrapping GUI which can be leveraged to backend application with the diagnostic problem depending on the electronic health record.

5 Literature Review

1. HIV prophylaxis programs in Europe and their evaluation methods ([ECDC, 2024](#))
2. Integration of FHIR standards in healthcare interoperability ([FHIR Manifesto, 2024](#)).
3. Role of Medical Information Objects in electronic health records ([MIO42, 2024](#))
4. KBV Basis Profile as a regulatory board ([KBV Basic Profiles, 2024](#))

6 Methodology

6.1 Data Collection

The study will gather data from diverse sources to ensure a comprehensive understanding of PrEP usage and its impact. Secondary data will be collected from pharmacies, the statutory health insurance (SHI) system, and the FDZ Health Data Lab, providing a robust quantitative foundation. This will be complemented by the integration of service-specific data from HIV specialty care centers within the dāgna network, offering insights into the operational aspects of PrEP provision. To address the social and systemic barriers to PrEP adoption, qualitative data will be collected from communal bodies. This multi-faceted approach will enable a holistic evaluation of PrEP adoption, effectiveness, and accessibility within Germany's healthcare system.

6.2. Digital Framework Development

The technical implementation will utilize a robust programming stack, with Python and JavaScript powering the backend, and Express JS, HTML, and CSS driving the frontend for a responsive and user-friendly interface. A structured SQL database will manage data storage and retrieval, while JSON will serve as the primary protocol for efficient two-way data transmission between the frontend and backend, enabling real-time communication and interaction within the system. This cohesive approach ensures that the framework is both technically sound and scalable for future applications.

GUI OWNER'S MANUAL FOR BIO/MEDICAL INFORMATICS

1. Ensure the application is complete. The Web/Mobile based application should suffice the three-tranche GUI architecture.

2. Select technologies accordingly which helps you to commit to user experience in bio/medical informatics which reflect the project's user-cases.
3. The long-term goal is to maximize the usability and effectiveness of such applications with a strict focus on each user rather than just the scale of the offerings. As the field grows, we anticipate that enhancing these GUIs will become more challenging, and commit to ensure that our improvements surpass those existing solutions providing real value to researchers.
4. Leveraging and bootstrapping --as a central key for creating these GUIs. Bootstrap elements should be publicly available which should be governed by a body.
5. The shorthand-version of languages to construct the frontend should be used which ensures easy Data Object Manipulation (DOM)
6. Animation based GUI must be avoided for any bio/medical informatics application except the visualization part of the data itself. This gives users a functional Command Line Interface-like environment in terms of memory escaping from graphics lag and operating system crashes which are headwinds for GUIs.
7. Develop a GUI which has a short learning curve for the user to adapt and specialize which in turn increases productivity.
8. Be consistent with the documentation which includes addition and removal of versions, code comments, formatting, visuals, installation procedures and troubleshooting guidelines.
9. GUI-based frameworks should be developed on two principles for computational optimization. a. Reuse components b. Reduce overhead.
10. Reduce technology improvement noise by sticking to the fundamental elements of frontend designing principles which ensures GUI longevity.

6.3 Data Analysis *[future works]*

The project will assess its impact through key metrics, including the adoption rates of PrEP and the incidence of HIV and sexually transmitted infections (STIs) among PrEP users. These metrics will provide crucial insights into the effectiveness and reach of PrEP services within the healthcare system. Advanced analytical tools will be employed for this purpose, with statistical modeling conducted using R and Python to identify trends and correlations. Visualization tools such as Tableau and Matplotlib can be used to present findings in an accessible and intuitive manner, enabling stakeholders to make data-driven decisions and optimize the provision of PrEP services.

7 Timeline

Phase	Duration	Deliverables
Literature Review	3 weeks	Annotated bibliography and draft background
Framework Development	2 months	Interoperable platform prototype
Data Analysis *	1 month	Statistical models and visualizations
Thesis Writing	1 month	Final thesis

*future work

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